SmolVLM: Redefining Small and Efficient Multimodal Models

Hugging Face × Stanford University

Compact VLM family (256M / 500M / 2.2B) built for edge inference
Sub-1GB RAM multimodal inference with competitive benchmark performance
Core innovations: compute allocation, pixel-shuffle compression, context scaling

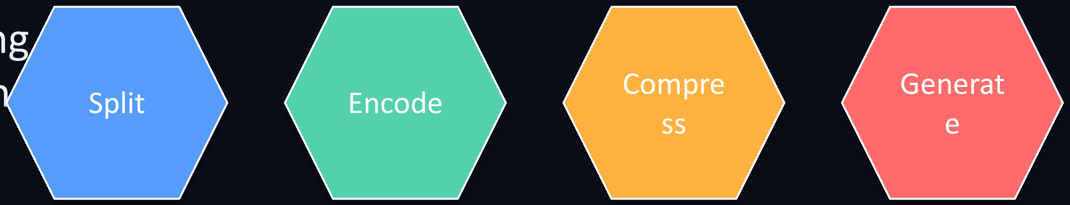
Image Splitting → SigLIP Encoder → Pixel Shuffle → Linear Projection → SmolLM2

Large VLMs Are Powerful but Impractical for Edge Deployment

State-of-the-art multimodal models often require server-class memory and compute.
Edge constraints: battery, thermals, memory bandwidth, and privacy-by-design requirements.
Example RAM at batch=1: SmolVLM-256M = 0.8GB vs Molmo-A1B-7B = 27.7GB.
Goal: retain competitive performance while enabling local, real-time inference.

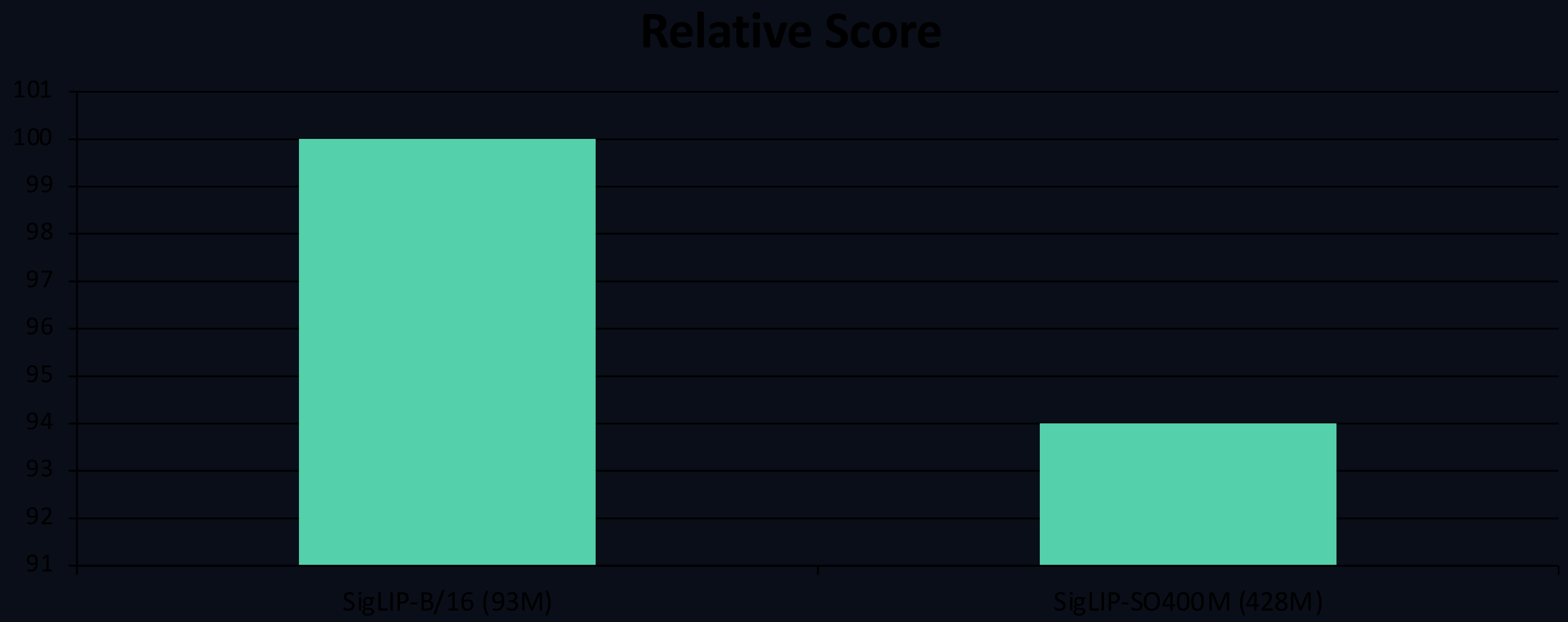
SmolVLM Architecture: Image Splitting, Pixel Shuffle, and SmolLM2 Backb

1) Split high-resolution image into sub-images to preserve detail.
2) Encode patches with SigLIP vision encoder.
3) Apply pixel shuffle token compression before language modeling
4) Project visual tokens to SmolLM2 space for multimodal decodin



Smaller Vision Encoders Outperform Larger Ones in Compact VLMs

Finding: for small LMs, balanced parameter allocation beats oversized vision towers.
93M SigLIP-B/16 outperforms 428M SigLIP-SO400M in smallest-model setting.



Aggressive Pixel Shuffle (r=4) Cuts Visual Tokens 16× for Small Models

Pixel shuffle rearranges spatial features to reduce sequence length before the LM.
At shuffle factor r=4, visual token count is reduced by 16×.
Small VLMs benefit most from r=4; larger variants often prefer r=2.

Extending Context from 2K to 16K Tokens Unlocks Higher Image Resolutio

Finding: extending context length materially improves multimodal performance.
RoPE base adjusted from 10K to 273K to support longer sequences.
Fine-tuning uses mixed short/long-context data for stability and transfer.
2.2B variant uses 16K context; 256M/500M variants use 8K context.

2K → 4K → 8K → 16K

SmolVLM Achieves Competitive Performance at Sub-1GB to 4.9GB RAM

Model	Params	Avg Score	RAM (GB, b=1)
SmolVLM-256M	256M	44.0%	0.8
SmolVLM-500M	500M	51.0%	1.2
SmolVLM-2.2B	2.2B	59.8%	4.9
InternVL2-2B	2B	Best Efficient OS baseline	n/a
Molmo-A1B-7B	7B	Best Efficient OS baseline	27.7

SmolVLM-256M also reported to surpass the ~300× larger Idefics-80B.

Real-Time Inference on iPhones: Up to 80 Tokens/Second on Apple Silicon

HuggingSnap runs fully on-device: photos/video processed locally (no cloud dependency).
SmolVLM-256M reaches ~80 tok/s on Pro M1 Max 64GB; ~55 tok/s on MacBook Air M3 16GB.
At batch size 64, 256M reaches nearly 1000 tok/s on high-end Apple Silicon.
Throughput scales roughly linearly with batch size for the smallest model.

Key Takeaways: Design Principles for Efficient Multimodal Models

Balance encoder and language-model capacity instead of maximizing one side.
Use aggressive token compression (r=4) for smallest models to control sequence cost.
Scale context length (8K–16K) with RoPE adaptation to support higher image resolution.
Sub-1GB multimodal inference is practical with careful architecture co-design.
Open release of models, data, and app accelerates reproducible edge-AI research.